

Compound Sustainability Stress Index (SSI)

Complete Reference Guide: Abstract, Formulas, & Data Pipeline

Project: Spatiotemporal Modeling of Compound Environmental Stress

Output Artifact: Publications & Submissions Package

Status: Approved by User

Date: August 2026

Google DeepMind Advanced Agentic Coding Project

Abstract Submission: Compound Sustainability Stress Index (SSI)

Title: High-Resolution Spatiotemporal Modeling of Compound Sustainability Stress in Rapidly Urbanizing Cities

Abstract: Rapid urbanization and climate change expose cities to compounding environmental risks. While various specialized indices evaluate isolated environmental or social hazards, a single, all-inclusive metric for compound urban stress remains absent. We present the Compound Sustainability Stress Index (SSI), a spatiotemporal framework that evaluates multi-dimensional urban stress. By mapping heterogeneous climate, satellite, air quality, urban form, and demographic data onto a uniform H3 hexagonal grid, the index calculates five distinct stress indicators: heat, water, air pollution, vegetation degradation, and urban vulnerability. These indicators are combined using Principal Component Analysis (PCA) to derive a dynamic composite score. Sourced and validated across major Indian cities (e.g., Mumbai), the pipeline identifies localized environmental hotspots and seasonal stress cycles, which peak during winter pollution and pre-monsoon dry phases. Furthermore, a clustering analysis groups urban sectors into six distinct policy-relevant archetypes, distinguishing between heat-dominated and vulnerability-dominated risks. Our results demonstrate that high-resolution, grid-scale composite modeling can identify spatial inequalities and compounding hazards that macro-level indicators overlook. Ultimately, the SSI framework provides urban planners and municipal authorities with an actionable decision-support tool to design and prioritize targeted, localized resilience interventions in rapidly developing metropolitan areas.

Keywords: Urban Resilience, Spatial-Temporal Modeling, Composite Indicators, H3 Hexagonal Grid, Environmental Stress, Decision Support Systems.

Abstract Metrics & Verification

- **Word Count:** 189 words (excluding Title, Keywords, and metadata headers; meets the < 200 words constraint)
- **Structure:** Self-contained, covers aim, methods, results, and conclusions.
- **Abbreviations Used:** SSI (defined as *Compound Sustainability Stress Index*), PCA (defined as *Principal Component Analysis*).

Mathematical Formulations in the Compound Sustainability Stress Index (SSI)

-- Plain Text Reference

This document provides a clean, plain-text reference of all formulas used throughout the Compound SSI pipeline. Every formula has been written using standard keyboard math notation (e.g. * for multiplication, / for division, ^ for powers) so it renders correctly and is fully readable in any editor.

1. Heat Stress Index (HSI)

The Heat Stress Index measures biometeorological heat risk by combining ambient perceived temperature with satellite-observed land skin temperatures.

Formula 1.1: Relative Humidity (RH) via Magnus-Tetens Approximation

Relative Humidity is calculated from the mean 2m air temperature (T, in degrees Celsius) and 2m dewpoint temperature (Td, in degrees Celsius):

$$RH = 100 * [\exp((17.625 * Td) / (243.04 + Td))] / [\exp((17.625 * T) / (243.04 + T))]$$
 Note: The calculated RH is clamped to stay strictly between 0 and 100 percent. The term "exp(x)" means raising the constant e (approx. 2.718) to the power of x.

- **Why we use it:** RH is critical because high humidity prevents the evaporation of sweat, which is the human body's primary cooling mechanism. The Magnus-Tetens approximation is the standard meteorological method for calculating saturation water vapor pressure under normal temperature conditions.

Formula 1.2: Perceived Temperature via Steadman/Rothfusz Heat Index (HI)

If temperature is greater than 26 deg C and relative humidity is greater than 40%, the perceived temperature feels hotter than the actual thermometer reading. It is calculated using the Rothfusz regression equation:

$$HI = -8.78469 + (1.61139411 * T) + (2.338549 * RH) - (0.14611605 * T * RH) - (0.01230809 * (T^2)) - (0.01642482 * (RH^2)) + (0.002211732 * (T^2) * RH) + (0.00072546 * T * (RH^2)) - (0.000003582 * (T^2) * (RH^2))$$
 Note: T^2 means temperature squared, and RH^2 means relative humidity squared.

If conditions do not meet the warm/humid threshold (cooler than 26 deg C or dry), it defaults to: $HI = T$

- **Why we use it:** Simple ambient temperature underestimates actual heat stress on human health. The Rothfusz equation is the standard US National Weather Service model that quantifies how hot the air actually feels by accounting for humidity-induced heat retention.

Formula 1.3: Blended Heat Index (Heat_raw)

We blend the ambient perceived Heat Index (HI) with Satellite Land Surface Temperature (LST, in degrees Celsius):

$$Heat_raw = (w_hi * HI) + (w_lst * LST)$$
 Note: The weights default to $w_hi = 0.5$ and $w_lst = 0.5$.

- **Why we use it:** HI measures the air temperature felt in the shade, while LST captures radiative land skin surface heat (the urban heat island effect, caused by concrete and asphalt absorbing solar heat). Blending both accounts for ambient atmospheric conditions and direct ground surface radiative exposure.

Formula 1.4: Baseline-Normalized HSI (HSI)

To isolate unusual anomalies, we calculate the z-score relative to the calendar-month baseline:

$$\text{HSI} = \text{Max}(0, (\text{Heat_raw} - \text{Mean_Baseline_for_Month}) / \text{StdDev_Baseline_for_Month})$$
Note: "Mean_Baseline_for_Month" and "StdDev_Baseline_for_Month" are the average and standard deviation of Heat_raw for that specific calendar month (e.g. all Januaries) during the historical baseline period. "Max(0, x)" means if the result is negative, it is set to 0.

- **Why we use it:** Absolute temperatures vary naturally by season. Evaluating z-scores per calendar month ensures that we measure anomalous deviations (e.g. an unseasonably hot winter day) rather than standard summer seasonality.

2. Water Stress Index (WSI)

The Water Stress Index captures agricultural, hydrological, and meteorological drought by balancing water demand (evapotranspiration) against water supply (precipitation and soil moisture).

Formula 2.1: Thornthwaite Potential Evapotranspiration (PET)

Potential Evapotranspiration (PET, in mm/month) is calculated as:

$$\text{Unadjusted_PET} = 16 * [(10 * T_{\text{clamped}}) / \text{Heat_Index}]^a$$
 where: * T_{clamped} is the mean monthly temperature, clamped to a minimum of 0 deg C. * Heat_Index is the sum over all 12 months of: $(\text{Monthly_Temp} / 5)^{1.514} * a = (6.75 * 10^{-7} * (\text{Heat_Index}^3)) - (7.71 * 10^{-5} * (\text{Heat_Index}^2)) + (0.01792 * \text{Heat_Index}) + 0.49239$
 The final adjusted PET accounts for latitude and calendar month length: $\text{PET} = \text{Unadjusted_PET} * (\text{Daylight_Hours} / 12) * (\text{Days_in_Month} / 30)$
(Daylight_Hours is computed from latitude and solar declination).

- **Why we use it:** Evapotranspiration represents the atmosphere's demand for water. The Thornthwaite method is a globally recognized, temperature-driven model that provides reliable PET estimates without requiring wind or solar radiation variables, which are often unavailable or noisy at local municipal scales.

Formula 2.2: Precipitation Deficit (Precip_deficit)

We compute the negative anomaly of monthly precipitation (Precipitation, in mm) relative to the calendar month baseline:

$$\text{Precip_deficit} = \text{Max}(0, (\text{Mean_Precip_Baseline_for_Month} - \text{Precipitation}) / \text{StdDev_Precip_Baseline_for_Month})$$

- **Why we use it:** Drought is defined relative to local climate norms. This scales rainfall deficits according to the typical variance of that specific month (e.g., monsoon months vs. dry winter months).

Formula 2.3: Soil Moisture Deficit (Soil_deficit)

Evaluated as the deviation below the historical baseline's 25th percentile:

$$\text{Soil_deficit} = \text{Max}(0, (\text{Baseline_25th_Percentile_for_Month} - \text{Soil_Moisture}) / \text{Baseline_25th_Percentile_for_Month})$$
Note: Soil_Moisture is the soil water volume fraction (m³/m³).

- **Why we use it:** Precipitation deficits take time to affect the ground. Soil moisture deficits represent active agricultural and hydrological drought. Comparing to the 25th percentile (first quartile) flags dry conditions that actively threaten vegetation.

Formula 2.4: Combined WSI (WSI)

The deficits are blended and normalized:

$$\text{Water_raw} = (w_{\text{precip}} * \text{Precip_deficit}) + (w_{\text{soil}} * \text{Soil_deficit})$$

$$\text{WSI} = \text{Max}(0, (\text{Water_raw} - \text{Mean_Baseline_for_Month}) / \text{StdDev_Baseline_for_Month})$$

- **Why we use it:** Blending precipitation and soil moisture captures both immediate meteorological drought (lack of rain) and delayed hydrological drought (dry soil) in a single unified index.

3. Pollution Exposure Index (PEI)

The Pollution Exposure Index measures chronic and acute exposure to PM2.5 air pollution relative to public health standards.

Formula 3.1: WHO Exceedance Component (WHO_exceedance)

Calculated as the fraction by which local PM2.5 concentrations exceed the WHO annual guideline threshold (15 micrograms per cubic meter):

$$\text{WHO_exceedance} = \text{Max}(0, (\text{PM2.5} - 15.0) / 15.0) \quad \text{WHO_exceedance_Normalized} = \text{MinMax_Normalize}(\text{WHO_exceedance})$$

- **Why we use it:** The World Health Organization sets 15 ug/m³ as a critical annual exposure threshold. Using a ratio-based approach ensures that areas with extreme pollution are penalized proportionally to the public health risk they carry.

Formula 3.2: Temporal Anomaly Component (Pol_anomaly)

The baseline-normalized z-score isolates local seasonal anomalies (e.g. stubble burning or winter temperature inversions):

$$\text{Pol_anomaly} = \text{Max}(0, (\text{PM2.5} - \text{Mean_PM2.5_Baseline_for_Month}) / \text{StdDev_PM2.5_Baseline_for_Month})$$

- **Why we use it:** A city might have high baseline pollution, but sudden acute spikes represent severe episodic hazards (e.g., smog events). Blending both ensures the index penalizes both chronic baseline exposure and acute spikes.

Formula 3.3: Combined PEI (PEI)

$$\text{PEI} = (0.5 * \text{WHO_exceedance_Normalized}) + (0.5 * \text{Pol_anomaly})$$

- **Why we use it:** Equates long-term systemic health risk (WHO limits) and short-term atmospheric episodes (z-score anomaly) to form a balanced assessment of air pollution stress.

4. Vegetation Degradation Index (VDI)

VDI monitors changes in green cover health, filtering out natural seasonal variations.

Formula 4.1: Inverted NDVI Z-Score (VDI)

NDVI values (0 to 1) are compared to historical calendar-month baselines, inverted, and floored at 0:

$$\text{NDVI_Z_Score} = (\text{NDVI} - \text{Mean_NDVI_Baseline_for_Month}) / \text{StdDev_NDVI_Baseline_for_Month} \quad \text{VDI} = \text{Max}(0, -1 * \text{NDVI_Z_Score})$$

- **Why we use it:** Vegetation loss is represented by a negative NDVI anomaly. Multiplying by -1 maps deficit to a positive stress scale. Computing the baseline per calendar month ensures that deciduous leaf-shedding during dry seasons is not falsely flagged as structural degradation.

5. Urban Vulnerability Index (UVI)

The Urban Vulnerability Index models structural and socio-economic sensitivity using static spatial indicators.

Formula 5.1: Physical/Structural Vulnerability (UVI_physical)

Integrates building density, building footprint fraction, road network density, and the absence of green spaces:

$$\text{UVI_physical} = (w1 * \text{Normalized_Building_Density}) + (w2 * \text{Normalized_Footprint_Fraction}) + (w3 * \text{Normalized_Road_Density}) + (w4 * (1 - \text{Normalized_Green_Space_Fraction}))$$
 Note: Sub-indicator weights (w1, w2, w3, w4) are derived dynamically using nested PCA, and values are minmax-normalized.

- **Why we use it:** High building footprint fractions and densities create heat-trapping microclimates and limit stormwater drainage. Road density indicates concrete coverage, while the inverse of green space captures the lack of local environmental buffers.

Formula 5.2: Socio-Economic Vulnerability (UVI_socio)

Combines the percentage of the population below the poverty line (BPL), slum residency fraction, elderly demographic fraction, and inverse literacy rates:

$$\text{UVI_socio} = (v1 * \text{Normalized_BPL_Percent}) + (v2 * \text{Normalized_Slum_Percent}) + (v3 * \text{Normalized_Elderly_Percent}) + (v4 * (1 - \text{Normalized_Literacy_Percent}))$$

- **Why we use it:** Socio-economic factors dictate a community's adaptive capacity. Poorer populations in slums lack air conditioning, insulation, and water storage, making them significantly more vulnerable to the same physical stress.

Formula 5.3: Combined UVI (UVI)

$$\text{UVI} = (0.4 * \text{UVI_physical}) + (0.6 * \text{UVI_socio})$$

- **Why we use it:** The 40/60 split prioritizes socio-economic adaptive capacity over physical form, reflecting established climate vulnerability frameworks (like the IPCC vulnerability formulation) where human susceptibility is the dominant driver of actual risk.

6. Composite Sustainability Stress Index (SSI)

The composite score aggregates the 5 stress indicators into a single, spatial-temporal compound hazard index.

Formula 6.1: PCA Eigenvector Weighting

The 5 indicators are standardized (mean = 0, variance = 1) to form a matrix. Principal Component Analysis is run to find the first principal component (PC1) eigenvector loadings. The loadings of PC1 define the indicator weights:

$$\text{Weight}_i = \text{Absolute_Value}(\text{Eigenvector_Loading}_i) / \text{Sum of Absolute_Values}(\text{All_5_Eigenvector_Loadings})$$

If the data has too few observations (fewer than 30 rows), the pipeline falls back to equal weights:

$$\text{Weight}_i = 0.20 \text{ (for all 5 indicators)}$$

- **Why we use it:** Using fixed arbitrary weights overlooks the fact that different cities have different primary drivers of environmental stress. PCA dynamically determines weights based on the actual variance and co-occurrence of hazards in the city, ensuring the index is mathematically objective and tailored to local conditions.

Formula 6.2: Normalized Composite SSI (SSI)

$$\text{SSI_raw} = (w_{\text{heat}} * \text{HSI}) + (w_{\text{water}} * \text{WSI}) + (w_{\text{pollution}} * \text{PEI}) + (w_{\text{veg}} * \text{VDI}) + (w_{\text{vuln}} * \text{UVI})$$

$$\text{SSI} = (\text{SSI_raw} - \text{Min_SSI_raw}) / (\text{Max_SSI_raw} - \text{Min_SSI_raw})$$

- **Why we use it:** Minmax normalization maps the final composite score onto a strict 0-to-1 scale, making it simple to interpret and compare across different locations and time periods.

Sustainability Stress Index (SSI) -- Data Pipeline & Preprocessing Guide

This document details the architecture of our data pipeline, covering the specifications of all raw data sources, data types, temporal scopes, spatial alignment, and the exact normalization procedures executed before calculating the Compound SSI.

1. Raw Data Sources & Specifications

The pipeline integrates five distinct raw data layers, each with its own structure and acquisition method.

Data Source	Specific Variables	Format	Data Type	Spatial Resolution	Temporal Scope	Source Platform
ERA5 Climate	2m Air Temperature, 2m Dewpoint, Total Precipitation, U/V Wind Components, Solar Radiation, Soil Moisture NDVI	.nc (NetCDF)	Raster / Grid	0.1 degree (~11 km)	2015 - 2025 (Hourly/Monthly)	Copernicus Climate Data Store (CDS)
Satellite Imagery	(Greenness Index), LST (Land Surface Temperature), Built-Up fraction	.tif (GeoTIFF)	Raster	NDVI: 10m (Sentinel-2); LST: 1km (MODIS)	Monthly composite frames	Microsoft Planetary Computer STAC
Urban Form	Building footprints, road network polylines, green space/park boundaries	.gpkg (GeoPackage)	Vector (Polygons Polylines)	Sub-meter & vector precision	Static (Current snapshot)	OpenStreetMap (OSM) via Overpass API
Air Quality	PM2.5 concentrations	.tif (GeoTIFF) or procedurally generated	Raster	0.01 degree (~1 km)	Monthly	NASA SEDAC (with Synthetic fallback)
Demographics	Slum %, Below Poverty Line %, Elderly population %, Literacy rate %	Tabular or spatial attributes .csv	Numeric Float/Int	District/Ward level	Decadal Census / Local priors	Census of India / Synthetic priors

2. Spatial & Temporal Harmonization

Before any mathematical calculations occur, all raw files are mapped to a uniform spatial-temporal coordinate system.

```
+-----+ +-----+ +-----+ +-----+
+-----+ | ERA5 NetCDF | | Satellite TIF | | OSM Vector | |
Demographics | | (11km Climate) | | (10m/1km Grid) | | (Building/Roads) | |
(Tabular) | +-----+ +-----+ +-----+ +-----+ +-----+
+-----+ +-----+ | | | v v v v Centroid Interpolation Zonal Statistics
(Mean) Spatial Buffers & Areas Centroid Mappings | | |
+-----+ +-----+ +-----+ +-----+ +-----+ +-----+ | v
Spatiotemporal Skeleton Join [H3 Cell Index (Res 8) x Calendar Month] | v
Unified Panel Parquet
```

Step 2.1: Spatial Grid Slicing (H3 Grid)

- We generate a contiguous grid of **H3 hexagonal cells at Resolution 8** covering the target city's bounding box.
- A Resolution 8 cell has an average area of **0.74 square kilometers** and an edge length of **461 meters**. This resolution is selected because it represents a standard walking neighborhood scale (pedestrian catchment area).

Step 2.2: The Spatiotemporal Skeleton

- The pipeline constructs a master temporal table containing the Cartesian product of all active H3 cell indexes and every calendar month within the study period (e.g. 120 months for 10 years).
- For a city like Mumbai with 2,501 hex cells monitored over 12 months, this creates a unified master skeleton of exactly **30,012 observation rows**.

Step 2.3: Data Mapping & Aggregation onto the Grid

Raw datasets are mapped to this skeleton using specific geospatial routines: * **Climate Grids (NetCDF)**: The centroid coordinates (Latitude/Longitude) of each H3 hex cell are used to query and spatially interpolate climate values from the 11km resolution ERA5 grid. * **Satellite Imagery (Rasters)**: For each monthly LST and NDVI GeoTIFF, zonal statistics are calculated. The pipeline computes the average pixel value falling within the geographic boundaries of each H3 hexagon using polygon rasterization. * **Urban Form (Vector polygons)**: * **Buildings**: The pipeline calculates building density (count of building centroids per H3 cell) and building footprint fraction (sum of building footprint polygon areas intersecting the H3 cell divided by the total area of the H3 cell). * **Roads**: Calculated as road density (total length of road polylines inside the H3 cell in kilometers divided by the H3 cell area in square kilometers). * **Green Spaces**: Sourced from OSM parks and gardens; calculated as the fraction of H3 cell area covered by green spaces. * **Demographics**: Census boundaries are intersected with H3 centroids. The demographic values (slum %, poverty rate) of the intersecting census tract are assigned to that H3 cell.

3. Data Preprocessing & Normalization

Once the raw attributes are successfully merged onto the H3 spacetime skeleton, they undergo structured normalizations before indicator formulas are applied.

Step 3.1: Temporal Gap Filling (Cloud Correction)

- Satellite imagery (especially NDVI) often suffers from cloud cover gaps.
- The pipeline applies a localized temporal linear interpolation (forward-fill and backward-fill) per H3 cell to reconstruct missing monthly values. If a cell remains missing, it falls back to the city-wide

calendar-month historic mean.

Step 3.2: Static Indicator Min-Max Normalization

Static indicators (demographics and urban form density) are normalized to a uniform $[0, 1]$ range to represent relative spatial exposure across the city: $\text{Normalized_Value} = (\text{Raw_Value} - \text{Min_City_Value}) / (\text{Max_City_Value} - \text{Min_City_Value})$. If a higher variable value represents a *lower* stress level (e.g. green space coverage or literacy rates), the normalization is inverted so that higher scores consistently represent higher stress: $\text{Inverted_Normalized_Value} = 1 - \text{Normalized_Value}$

Step 3.3: Dynamic Indicator Calendar-Month Baseline Z-Score Normalization

Climatic and satellite factors (Temperature, Precipitation, Soil Moisture, NDVI, PM2.5) are highly seasonal. Normalizing them against a global mean would falsely flag standard winters or summers as anomalies. To resolve this, we normalize dynamic features using a **monthly baseline z-score** derived from a historical reference period (typically the first 3 to 10 years of data):

1. **Baseline Calculations:** For every H3 cell, we calculate the historic average (Mean_Baseline) and standard deviation (StdDev_Baseline) for each of the 12 calendar months individually (e.g., the average temperature of all Januaries in the baseline period).
2. **Z-Score Mapping:** Each monthly observation is evaluated against its corresponding month's baseline parameters: $\text{Z_Score} = (\text{Raw_Value} - \text{Mean_Baseline_for_Month}) / \text{StdDev_Baseline_for_Month}$
3. **Stress Flooring:** Since we are interested in elevated stress, negative z-scores (indicating conditions cooler, wetter, or cleaner than average) are floored at 0 using a maximum function: $\text{Stress_Score} = \text{Max}(0, \text{Z_Score})$
4. **Vegetation Inversion:** Because higher greenness (NDVI) indicates healthier vegetation (lower stress), the NDVI z-score is multiplied by -1 so that a negative vegetative anomaly represents positive degradation stress: $\text{VDI_Score} = \text{Max}(0, -1 * \text{NDVI_Z_Score})$

These preprocessed, normalized indicators are then passed directly into the final feature scoring and composite index calculations.